



SUNBREEZE OFF-GRID GUIDES

THE OFF-GRID BATTERY HANDBOOK

How to charge, store, and get years out of the battery in your SunBreeze Power Bank — and any lithium power station you run off solar.

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WHY BATTERIES DIE EARLY

Almost every "dead" power bank isn't dead — it's degraded. Lithium batteries (including the LiFePO₄ and Li-ion cells used in solar power banks) don't fail suddenly; they lose capacity gradually every time they're charged and discharged, and how you treat them between uses decides whether that loss happens over 300 cycles or 3,000. This guide is the difference.

Two numbers matter more than anything else: **heat** and **state of charge at rest**. Everything below comes back to those two things.

HOW YOUR BATTERY ACTUALLY WORKS

A solar power bank has three parts working together: the solar panel (collects energy), the charge controller (regulates how that energy enters the battery), and the battery itself (stores it as chemical energy). Modern power banks, including SunBreeze's, use a built-in Battery Management System (BMS) that protects against overcharge, over-discharge, short circuits, and excess heat — but a BMS manages risk, it doesn't undo bad habits.

Li-ion vs. LiFePO₄

If your unit uses standard lithium-ion cells, expect roughly 300–500 full charge cycles before capacity drops meaningfully. If it uses LiFePO₄ (lithium iron phosphate) cells, expect 2,000–3,000+ cycles — LiFePO₄ is more heat-tolerant and safer, which is why it's becoming the standard for portable power stations. Check your unit's spec sheet or product label to know which chemistry you're working with, since the storage guidance below is easier on LiFePO₄ than on standard Li-ion.

THE CHARGING RULES THAT ACTUALLY EXTEND LIFE

1. Avoid the extremes

Letting a lithium battery sit at 0% or 100% for long stretches is the single biggest driver of early degradation. Charging to 100% occasionally is fine; living at 100% is not.

Rule of thumb: for daily or weekly use, keep the battery roughly between 20% and 90%. Full charge and full discharge are for the days you actually need the range — not the default state.

2. Charge at moderate speed when you can

Fast charging generates more heat, and heat is what actually damages the cells (the current itself is less of a factor than the temperature rise it causes). If you're topping off overnight or have no urgency, a slower charge via the solar panel or a standard USB charger will run cooler than a high-wattage fast charger.

3. Let it cool before and after charging

Don't charge a power bank that's been sitting in direct sun on a dashboard or in a hot trunk. Let it come down to something close to room temperature first. The same goes for charging immediately after heavy use (like running the SunBreeze Fan on high for hours) — give it 15–20 minutes to cool down.

4. Solar charging specifics

- **Angle the panel toward the sun**, not just toward the sky. Direct, perpendicular sunlight charges dramatically faster than a flat or shaded angle.
- **Clean the panel** — dust, pollen, and fingerprints measurably cut charging efficiency. A dry microfiber cloth once a week during heavy use is enough.
- **Partial shade kills output disproportionately** — even a small shadow across one corner of the panel can drop total output far more than the shaded area would suggest, because of how solar cells are wired in series. Reposition rather than accept partial shade.
- **Overcast still works, just slower.** Solar panels charge on daylight, not direct sun alone — expect 10–25% of peak output on a cloudy day rather than zero.

STORAGE: THE PART EVERYONE SKIPS

If you're not using the power bank for more than a few weeks — end of camping season, between trips, in a bug-out bag, in a garage — storage habits matter more than charging habits.

Storage length	Charge level to store at	Check-in frequency
1–4 weeks	Any level is fine	None needed
1–6 months	40–60%	Every 2–3 months, top back to 50% if low
6+ months / long-term emergency stock	50%	Every 3–4 months

Never store at 0%. A lithium cell left fully drained for months can drop into a "deep discharge" state that some BMS units won't recover from. If you know you won't touch it for a season, charge it to roughly half before you put it away.

Where to store it

Room temperature, out of direct sun, away from a hot garage wall or a cold, damp shed. Lithium cells degrade faster at both temperature extremes — the sweet spot is roughly 50–77°F (10–25°C). If you're storing it in a vehicle or a bug-out kit, that's a real-world compromise; just try to keep it out of the trunk in summer and check the charge level more often.

SAFETY BASICS

- Never puncture, crush, or disassemble the unit. If it's swollen, misshapen, or the casing feels warped, stop using it and don't attempt to charge it — recycle it at a battery drop-off point.
- Don't charge it somewhere it can't dissipate heat — a sealed pack, under a pillow, or buried in a bag. Give it open air.
- If it gets wet, dry it fully before charging or using it. SunBreeze Power Banks are weather-resistant, not submersible.
- Use the cable and adapter it came with, or a reputable equivalent, for the fastest and safest charge rate.

TROUBLESHOOTING

Symptom	Likely cause	What to try
Won't hold a charge like it used to	Normal cycle wear, or long storage at 0% or 100%	Run it through 2–3 full charge/discharge cycles to recalibrate the gauge; if capacity is still low, that's expected degradation over time
Solar panel charging very slowly	Poor angle, partial shade, or a dirty panel	Reposition directly at the sun, clean the panel, confirm no shadow crosses it
Won't turn on at all	Deep discharge or a tripped protection circuit	Plug into a wall charger for 30–60 minutes before trying again — the BMS sometimes needs a real charging source to reset, not just solar

Feels hot while charging	Normal under fast charge or direct sun; concerning if very hot to the touch	Move out of direct sun, switch to a slower charge source; if it stays hot after disconnecting, stop use and contact support
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YOUR QUICK-REFERENCE CHECKLIST

- ☐ Keep daily-use charge between 20–90% when possible
- ☐ Let it cool before charging after heavy use or sun exposure
- ☐ Angle solar panels directly at the sun; wipe them down weekly in heavy use
- ☐ Store at 50% charge, not 0% or 100%, for anything longer than a month
- ☐ Check stored units every 2–3 months and top up if needed
- ☐ Keep it between 50–77°F (10–25°C) when stored
- ☐ Never store or charge a swollen or damaged unit

The Off-Grid Battery Handbook — a SunBreeze bonus guide. General care guidance for lithium-based portable power banks; always defer to your unit's specific manual for chemistry, voltage, and warranty-specific instructions.
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